

KUNAL SHETTY

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LeetCode: 600+ problems solved | Contest Rating: 2300+ | Codeforces: Specialist

PROFESSIONAL SUMMARY

Full-stack engineer and solo builder focused on shipping AI-powered products end-to-end. Built production-grade systems spanning LLM orchestration, real-time infrastructure, semantic search, and hybrid local/cloud AI pipelines. Experienced in owning architecture, backend APIs, frontend experiences, deployment, and production systems from idea to launch.

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python, SQL, HTML, CSS

AI / ML: Generative AI, RAG, MCP, NLP, OCR, CNNs, Transformers

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, FastAPI, REST APIs, WebSockets, Socket.io

Cloud & DevOps: AWS EC2, Cloudflare R2, PostgreSQL, Firebase, GitHub Actions, CI/CD, Razorpay SDK

PROFESSIONAL EXPERIENCE

Classbot

Jul 2025 – Jan 2026

Full Stack Developer Intern | Mumbai, Malad (On-site)

- Engineered and optimized **20+ RESTful APIs** along with scalable SQL data models, implementing efficient query structures, **indexing** strategies that reduced average response times by **over 25%** and improved overall backend throughput
- Led frontend and backend improvements by designing responsive, cross-device dashboards for **300+ instructors** and refactoring legacy modules, resolving **30+ production issues**, enhancing system stability, and significantly reducing crash rates

Snapnotes

Jan 2026 – Present

Founder | Remote

- Architected and solo-shipped a full-stack **AI Edtech platform** across web and Android — **production API routes, 25,000+ lines of code**, and **30+ features** spanning notes, classrooms, live quizzes, a creator marketplace, and spaced repetition workflows.
- Integrated Gemini and Groq alongside locally deployed **Piper TTS** and **Whisper Base**, powering **7 distinct AI generation pipelines** including quiz generation, note summarization, semantic search via pgvector, and more — balancing cloud and local inference for cost and latency efficiency.
- Owned the complete infrastructure stack — **WebSocket-based** real-time multiplayer quiz rooms with live leaderboards and QR joining, **Razorpay** payment integration with **HMAC verification**, **Firebase push notifications**, **custom JWT auth** — deployed as a production-ready Android application.

KEY PROJECTS

Deimos — Autonomous AI Coding Agent [Github](#) | *Python, Groq, MCP, Shell, Agentic Harnessing, REPL*

- Engineered a terminal-native autonomous coding agent with a **multi-step agentic execution loop** (think→act→observe), orchestrating 10+ deterministic developer tools — file I/O, shell execution, AST-aware code generation — with **dynamic tool dispatch** and Groq Llama 3.3 as the inference backbone for sub-second response latency.
- Architected a **3-layer persistent memory hierarchy** (working, episodic, and semantic) backed by **vector-indexed Supabase storage**, enabling cross-session context reconstruction, automatic session distillation, and long-horizon project-aware reasoning across unbounded conversation depth.

Echo — Voice AI Finance Assistant [Link](#) | *Next.js, Groq, Speech-to-Text, Text-to-Speech, NLP*

- Designed an end-to-end **multimodal finance intelligence pipeline** — Whisper-based ASR ingestion, **zero-shot intent classification**, and structured entity extraction atomically persisting expense records under 2 seconds, with a context-aware **NLQ engine** surfacing **real-time TTS responses** and Wrapped-style breakdowns across 10+ expense dimensions.
- Architected a hybrid on-device/cloud inference model across a 2-layer processing architecture — routing low-latency voice capture and speech processing locally on-device while offloading financial reasoning and intent resolution to Groq's cloud inference layer, reducing per-session compute costs while maintaining sub-second conversational response times.

EDUCATION

Thakur College of Science and Commerce, Mumbai

Jul 2024 – Present

Bachelor of Science in Information Technology

Annasaheb Vartak College, Vasai

Jun 2023 – May 2024

Higher Secondary Certificate — Science Stream

Class 12: 89%

ACHIEVEMENTS & CERTIFICATIONS

- **NMIMS Cyber Cypher 5.0 - Hackathon Finalist:** Qualified Round 2 (Final Round) of 18-hour Hackathon at NMIMS Mukesh Patel School of Technology Management & Engineering, Mar 2026
- **India AI Impact Buildathon, HCL GUVI:** Competed among 40,000+ participants at the India AI Impact Summit 2026, Feb 2026
- **Avishkar Research Competition:** Presented research project in Engineering & Technology category at Thakur College intra-collegiate research competition, Sep 2025
- **IIT Bombay - Game Development Workshop:** Attended game development workshop at IIT Bombay, Dec 2025
- **Anthropic:** Claude Code in Action, AI Fluency: Framework & Foundations - Course Completion